SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 3 – Technical Presentation(5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 3: Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct

I **NIVEDITHA-192421440L** *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: NIVEDITHA

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Technical

Presentation

Q1. A shopping mall uses CCTV cameras that produce clear images during the day but dark and noisy images at night, making it difficult to identify people and objects.

Prepare a technical presentation demonstrating how filtering, noise removal, contrast enhancement, and sharpening techniques can be applied to improve the captured images. Present the sequence of image-processing operations required before object detection.

ANS:

Improving Night-Time CCTV Images

1. Problem Statement

- **Shopping mall CCTV cameras produce clear images during the daytime.**
- **At night, images become dark, blurry, and noisy.**
- **Poor image quality makes it difficult to identify people and objects.**
- **Image processing techniques improve image quality before object detection.**

2. Objectives

- **Improve the brightness of night-time images.**
- **Remove unwanted noise and blur where possible.**
- **Improve image contrast and object details.**
- **Increase the accuracy of object detection.**
- **Improve shopping mall security.**

3. Techniques Used

1. Filtering

- **Filtering reduces unwanted variations in an image.**
- **It improves the overall quality of CCTV images.**
- **Gaussian filtering can reduce certain types of noise.**
- **Proper filtering helps preserve important object details.**

2. Noise Removal

- **Noise appears as unwanted dots or grainy patterns.**
- **A median filter removes noise while preserving important edges.**
- **It makes people and objects easier to identify.**
- **It improves the quality of images captured in poor lighting.**

3. Contrast Enhancement

- **Contrast enhancement improves the difference between dark and bright areas.**
- **It makes objects more visible in dark images.**
- **Histogram equalization can improve image contrast.**
- **It helps reveal details that are difficult to see at night.**

4. Sharpening

- **Sharpening makes object boundaries and details clearer.**
- **It improves the visibility of edges.**
- **It helps distinguish people from the background.**
- **Excessive sharpening may increase unwanted noise.**

5. Object Detection

- **Object detection identifies people, bags, vehicles, and other objects.**
- **The enhanced image is given to the detection system.**
- **The system locates objects and identifies their categories.**

- **Better image quality can improve detection accuracy.**

4. Processing Sequence

- **Capture the CCTV image.**
- **Apply filtering to reduce noise.**
- **Remove unwanted noise.**
- **Enhance brightness and contrast.**
- **Sharpen important details.**
- **Perform object detection.**

5. Advantages

- **Improves night-time image quality.**
- **Helps identify suspicious activities.**
- **Reduces detection errors.**
- **Improves security monitoring.**
- **Supports faster identification of people and objects.**

Q2. A university is developing an automated document-scanning system using a camera, where captured documents may contain shadows, noise, uneven illumination, tilted pages, and blurred text.

Prepare a technical presentation explaining how filtering, thresholding, contrast enhancement, edge detection, and segmentation can be applied to improve the quality of the document images. Demonstrate the role of each technique in the processing pipeline.

ANS:

Improving Document Images for Automated Scanning

1. Problem Statement

- **A university uses cameras to scan documents automatically.**

- **Images may contain shadows, noise, and uneven illumination.**
- **Pages may be tilted, and text may be blurred.**
- **Image processing improves document quality and readability.**

2. Objectives

- **Remove noise and unwanted shadows.**
- **Improve text visibility.**
- **Detect document boundaries.**
- **Separate text from the background.**
- **Improve the accuracy of text extraction.**

3. Techniques Used

1. Filtering

- **Filtering removes unwanted noise from document images.**
- **Median filtering can reduce small noise particles.**
- **It makes text clearer and easier to process.**
- **Excessive filtering may remove thin text details.**

2. Thresholding

- **Thresholding converts grayscale images into black and white.**
- **It separates dark text from the light paper background.**
- **Global thresholding works well under uniform lighting.**
- **Adaptive thresholding is useful when lighting varies across the page.**

3. Contrast Enhancement

- **It improves the difference between text and paper.**
- **Faint letters become easier to read.**
- **It helps process faded or poorly illuminated documents.**
- **It improves the quality of scanned documents.**

4. Edge Detection

- **Edge detection identifies boundaries in an image.**
- **It helps locate the edges of the document.**
- **It supports page alignment and tilt correction.**
- **Clear edges make document processing easier.**

5. Segmentation

- **Segmentation divides an image into meaningful regions.**
- **It separates text, pictures, and background areas.**
- **It allows each region to be processed separately.**
- **It supports accurate text recognition.**

4. Processing Sequence

- **Capture the document using a camera.**
- **Apply filtering to remove noise.**
- **Enhance contrast.**
- **Detect document edges.**
- **Correct page tilt.**
- **Apply thresholding.**
- **Segment text and image regions.**
- **Use Optical Character Recognition (OCR) to extract text.**

5. Advantages

- **Improves document readability.**
- **Reduces scanning errors.**
- **Helps extract text accurately.**
- **Saves time in document management.**
- **Supports digital storage and searching.**

Q3. A fruit-processing company wants to automatically identify damaged apples from images captured on a conveyor belt, where shadows, reflections, different backgrounds, and brightness variations are present.

Prepare a technical presentation comparing global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation for this application. Demonstrate how a suitable technique or combination of techniques can improve fruit defect detection.

ANS:

Detecting Damaged Apples Using Image Segmentation

1. Problem Statement

- **A fruit-processing company wants to identify damaged apples automatically.**
- **Images may contain shadows, reflections, and different backgrounds.**
- **Brightness variations make defect detection difficult.**
- **Image segmentation helps separate apples and identify damaged regions.**

2. Objectives

- **Identify healthy and damaged apples.**
- **Separate apples from the background.**
- **Detect visible spots and damaged areas.**
- **Reduce manual inspection.**
- **Improve fruit quality and sorting accuracy.**

3. Segmentation Techniques

1. Global Thresholding

- **Uses a single threshold value for the entire image.**
- **Separates pixels based on their intensity.**
- **Works well when lighting is uniform.**

- **It may fail when shadows and brightness vary.**

2. Adaptive Thresholding

- **Uses different threshold values for different image regions.**
- **Handles uneven lighting better than global thresholding.**
- **Helps identify defects under shadows.**
- **It is useful when apples have different brightness levels.**

3. Edge-Based Segmentation

- **Detects object boundaries using intensity changes.**
- **Helps identify apple shapes and boundaries.**
- **It can locate some defects with clear edges.**
- **Reflections and unclear boundaries may reduce accuracy.**

4. Region-Based Segmentation

- **Groups similar pixels based on colour or intensity.**
- **Separates apples from the background.**
- **Helps identify regions with unusual colours.**
- **It works better when the fruit and background have different features.**

4. Proposed Method

- **Capture apple images using a camera.**
- **Apply filtering to remove noise.**
- **Enhance brightness and contrast.**
- **Use adaptive thresholding or region-based segmentation.**
- **Separate individual apples from the background.**
- **Extract colour, shape, and texture features.**
- **Identify damaged regions.**
- **Classify apples as healthy or damaged.**

5. Advantages

- **Detects damaged apples quickly.**
- **Reduces manual inspection.**
- **Improves fruit quality.**
- **Reduces waste.**
- **Increases production efficiency.**

Q4. An autonomous vehicle uses a front-facing camera to detect lane markings under different weather and lighting conditions, where shadows, damaged markings, vehicles, and rain may affect the captured images. **Prepare a technical presentation demonstrating how grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and region-of-interest techniques can be applied for lane detection. Explain the challenges associated with each processing stage.**

ANS:

Lane Detection in Autonomous Vehicles

1. Problem Statement

- **Autonomous vehicles use cameras to detect lane markings.**
- **Shadows, rain, and damaged markings affect image quality.**
- **Other vehicles may hide lane markings.**
- **Image processing helps vehicles identify lanes and move safely.**

2. Objectives

- **Detect lane markings accurately.**
- **Improve visibility under different lighting conditions.**
- **Reduce the effects of noise.**
- **Support safe and reliable driving.**

3. Techniques Used

1. Grayscale Conversion

- **Converts a colour image into grayscale.**
- **Reduces image complexity.**
- **Makes later processing easier.**
- **However, useful colour information may be lost.**

2. Filtering

- **Removes unwanted noise from road images.**
- **Reduces the effect of rain-related visual noise.**
- **Improves image quality before edge detection.**
- **Excessive filtering may remove important lane details.**

3. Contrast Enhancement

- **Improves the difference between lane markings and the road.**
- **Makes faded lane markings more visible.**
- **Helps in low-light conditions.**
- **Strong reflections may still cause errors.**

4. Edge Detection

- **Identifies boundaries and intensity changes.**
- **Helps locate lane markings.**
- **Supports the detection of straight and curved road boundaries.**
- **Damaged or faded markings may be difficult to detect.**

5. Thresholding

- **Separates bright lane markings from darker road surfaces.**
- **Converts images into simpler regions.**
- **Helps identify possible lane pixels.**
- **Shadows and headlights may create false detections.**

6. Region of Interest (ROI)

- **Selects the road area where lane markings are expected.**
- **Ignores unnecessary areas such as the sky and buildings.**
- **Reduces processing time.**
- **Incorrect ROI selection may miss distant or curved lanes.**

4. Processing Sequence

- **Capture the road image.**
- **Convert the image into grayscale.**
- **Apply filtering to reduce noise.**
- **Enhance contrast.**
- **Detect edges.**
- **Apply thresholding.**
- **Select the road region using ROI.**
- **Identify lane markings.**
- **Provide lane information to the vehicle control system.**

5. Advantages

- **Helps vehicles remain within lanes.**
- **Improves driving safety.**
- **Supports autonomous navigation.**
- **Reduces unnecessary processing.**
- **Helps vehicles respond to road conditions.**

Q5. A municipal corporation wants to classify waste such as plastic, paper, and metal using a camera installed above a conveyor belt. The captured images may contain overlapping objects, shadows, reflections, noise, and different object

orientations.

Prepare a technical presentation demonstrating a complete image-processing pipeline using preprocessing, enhancement, segmentation, feature extraction, and classification. Explain how the proposed pipeline can be applied to handle real-world variations and improve waste classification.

ANS:

Waste Classification Using Computer Vision

1. Problem Statement

- **A municipal corporation wants to classify plastic, paper, and metal waste.**
- **A camera captures waste moving on a conveyor belt.**
- **Images may contain overlapping objects, shadows, reflections, and noise.**
- **Computer vision helps automate waste classification and sorting.**

2. Objectives

- **Identify different types of waste.**
- **Separate objects from the background.**
- **Improve classification accuracy.**
- **Reduce manual labour.**
- **Support effective recycling.**

3. Techniques Used

1. Preprocessing

- **Preprocessing prepares images for further analysis.**
- **Resizing makes images suitable for the classification system.**
- **Filtering removes unwanted noise.**
- **It improves the consistency of input images.**

2. Image Enhancement

- **Improves brightness, contrast, and visibility.**

- **Helps handle shadows and uneven lighting.**
- **Makes object features easier to identify.**
- **Reduces the effect of poor image quality.**

3. Segmentation

- **Separates waste objects from the background.**
- **Identifies individual objects where possible.**
- **Helps process different waste materials separately.**
- **Overlapping objects may require additional segmentation methods.**

4. Feature Extraction

- **Extracts useful information from waste images.**
- **Colour helps distinguish some materials.**
- **Shape and size help identify bottles, sheets, and cans.**
- **Texture and surface appearance provide additional information.**
- **Extracted features are used for classification.**

5. Classification

- **Assigns waste objects to different categories.**
- **Machine learning models can classify plastic, paper, and metal.**
- **The system learns patterns from labelled training images.**
- **Correct classification supports automatic waste sorting.**

4. Proposed Method

- **Capture waste images using a conveyor-belt camera.**
- **Preprocess images and remove noise.**
- **Enhance brightness and contrast.**
- **Segment waste objects from the background.**
- **Extract colour, shape, size, and texture features.**
- **Classify objects as plastic, paper, or metal.**

- **Send classification results to the sorting mechanism.**
- **Monitor the results and improve the model using additional training data.**

5. Handling Real-World Variations

- **Overlapping objects: Use suitable segmentation methods to separate objects.**
- **Shadows: Apply contrast enhancement and adaptive thresholding.**
- **Reflections: Use suitable lighting and image preprocessing.**
- **Noise: Apply filtering before segmentation.**
- **Different orientations: Train the classifier using images captured from different angles.**
- **Different backgrounds: Use diverse training images to improve reliability.**

6. Advantages

- **Reduces manual waste sorting.**
- **Improves sorting speed and efficiency.**
- **Supports better recycling.**
- **Reduces classification errors.**
- **Handles large quantities of waste.**
- **Helps maintain a cleaner environment.**

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director

Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____